

■ AI/ML Internship – Day 34 Notes & Tasks

● Module 7: Introduction to Machine Learning & Types of ML

📌 Part 1: What is Machine Learning?

◆ Definition

👉 Machine Learning (ML) is a branch of Artificial Intelligence where computers learn patterns from data and make predictions without being explicitly programmed.

📌 Part 2: Simple Real-Life Examples

👉 Examples of ML:

- YouTube recommendations
 - Netflix suggestions
 - Face unlock in phones
 - Spam email detection
 - Google Maps traffic prediction
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📌 Part 3: Traditional Programming vs Machine Learning

Traditional Programming	Machine Learning
Rules are manually written	System learns from data
Fixed output	Predicts output
Less flexible	More intelligent

📌 Part 4: How Machine Learning Works?

◆ Basic Flow

Data → Training → Learning Patterns → Prediction

◆ Example

👉 Student marks prediction:

- Input = study hours
- Output = predicted marks

Machine learns relationship between them.

✦ Part 5: Theory Answers (IMPORTANT)

◆ 1. What is Machine Learning?

👉 Machine Learning is a method where computers learn from data and improve automatically.

◆ 2. Why Machine Learning is Important?

👉 ML helps:

- Automate tasks
 - Predict future outcomes
 - Analyze large data quickly
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◆ 3. What is a Dataset?

👉 Dataset is a collection of data used for training ML models.

◆ 4. What is Training?

👉 Training means teaching the ML model using data.

◆ 5. What is Prediction?

👉 Prediction means generating output based on learned patterns.

📌 Part 6: Types of Machine Learning

There are mainly 3 types:

1. Supervised Learning
 2. Unsupervised Learning
 3. Reinforcement Learning
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📌 Part 7: Supervised Learning

◆ Definition

👉 Model learns using labeled data.

👉 Labeled data means:

Input + correct output already available.

◆ Example

Study Hours Marks

2	40
4	70
6	90

👉 Model learns:

More study hours → more marks.

◆ Real-World Uses

- House price prediction
 - Spam detection
 - Student result prediction
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📌 Part 8: Unsupervised Learning

◆ Definition

👉 Model learns from unlabeled data.

👉 No correct output is given.

◆ Example

Customer grouping based on shopping habits.

◆ Real-World Uses

- Customer segmentation
 - Recommendation systems
 - Pattern detection
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📌 Part 9: Reinforcement Learning

◆ Definition

👉 Model learns through rewards and punishments.

◆ Example

👉 Robot learning to walk:

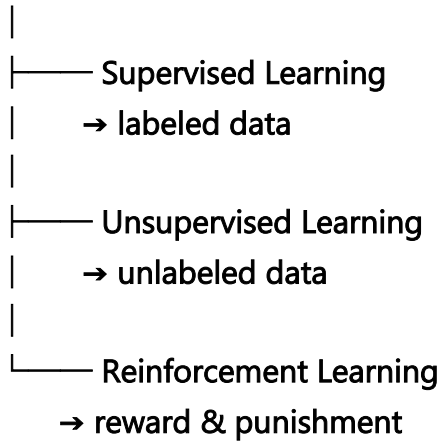
- Correct movement → reward
 - Wrong movement → penalty
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◆ Real-World Uses

- Self-driving cars
 - Robotics
 - AI game playing
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📌 Part 10: Visualization of ML Types

Machine Learning



📌 Part 11: Machine Learning Workflow

Collect Data



Clean Data



Train Model



Test Model



Prediction

📌 Part 12: Important ML Terms

Term	Meaning
Data	Information used for training
Feature	Input variable
Label	Correct output
Model	Learned system
Prediction	Final output

Part 13: Real-World Thinking

 ML is used everywhere today:

- Healthcare
- Banking
- Shopping apps
- Social media
- Security systems

 AI systems become smarter using ML.

Part 14: Why Python is Used in ML?

 Python is popular because:

- Easy syntax
- Large libraries
- Strong AI community

Libraries:

- NumPy
 - Pandas
 - Scikit-learn
 - TensorFlow
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Tasks for Day 34

Task 1: Theory (Write Answers)

1. What is Machine Learning?
2. Why ML is important?
3. What is dataset?

4. Difference between AI and ML
 5. Types of Machine Learning
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Task 2: Real-World ML Examples

1. List 5 ML applications
 2. Explain spam detection
 3. Explain recommendation system
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Task 3: Supervised Learning Practice

1. Create student marks dataset
 2. Identify features & labels
 3. Explain prediction logic
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Task 4: Unsupervised Learning Practice

1. Customer grouping example
 2. Pattern finding example
 3. Clustering example
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Task 5: Reinforcement Learning Practice

1. Robot learning example
 2. AI game example
 3. Reward/penalty explanation
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Task 6: ML Workflow

1. Draw ML workflow diagram
2. Explain each step

3. Give real-world example